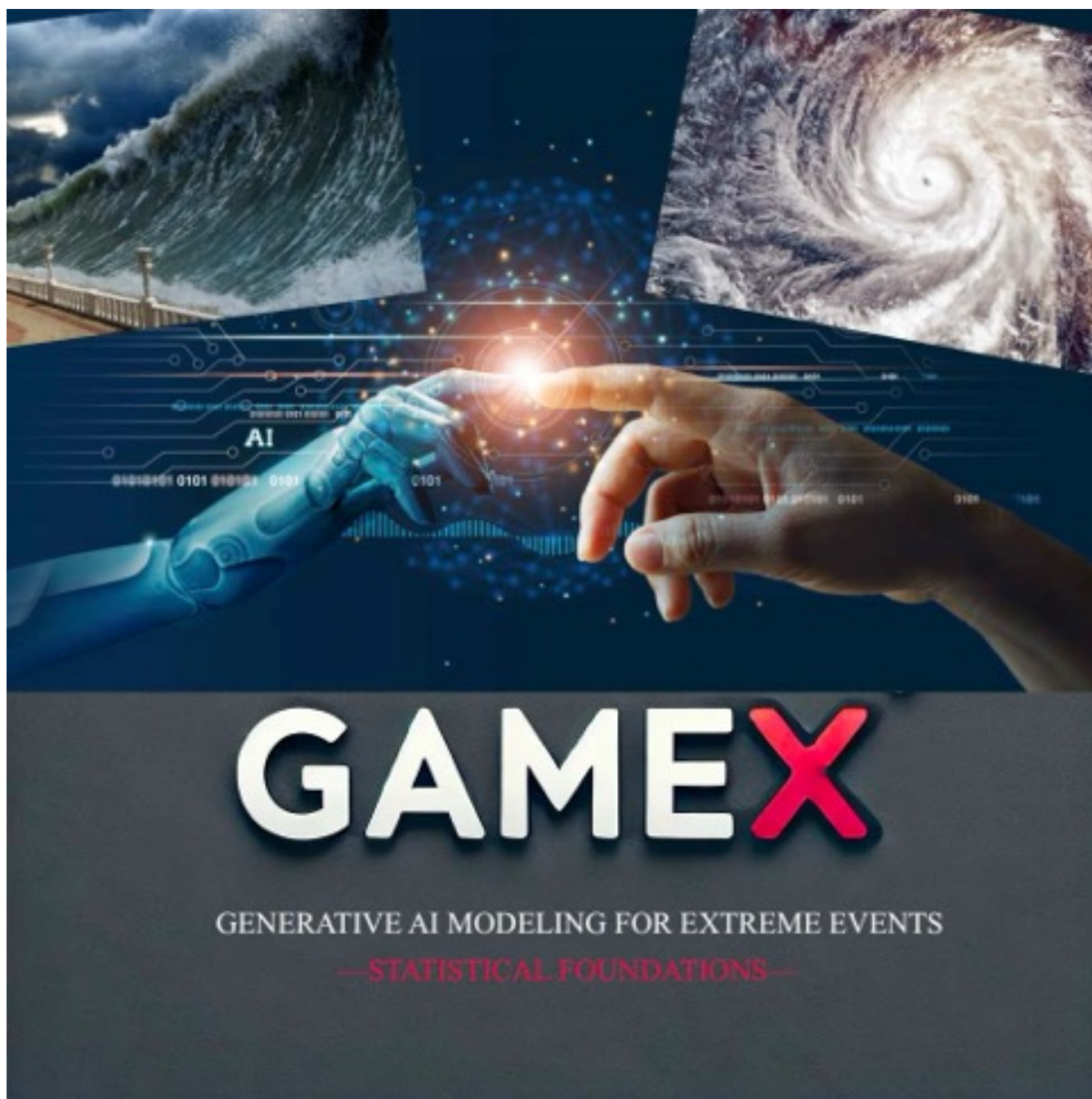


Edinburgh Summer School on Generative AI for Extremes



THE UNIVERSITY
of EDINBURGH



Provisional Programme—updated 2 September 2026
8–11 September 2026

<https://gamex-network.github.io/school/>

GL(EN)² (Glasgow–Lancaster–Edinburgh–Newcastle) Extremes Network

GAMEX Project Members: Luca Trapin • Miguel de Carvalho • Massimo Ventrucci • Ivo Bonfanti • Johnny Lee • Giorgia Ramponi
Till Freihaut • Brendan Murphy • Luis Gutierrez Inostroza • Clemente Ferrer

Chair

Professor Miguel de Carvalho

Chair of Statistics & Data Science

Elected Fellow of Generative AI Laboratory

Organisers

Johnny Lee (Edinburgh), Lambert De Monte (Edinburgh, KAUST), Jordan Richards (Edinburgh), Daniela Castro-Camilo (Glasgow), Mengran Li (Glasgow), Teng Wei Yeo (Glasgow), and Luca Trapin (Bologna).

Welcome

Building on the success of GAME 2025 (<https://edin.ac/4akncn7>) in Edinburgh, the first worldwide workshop on generative AI modelling for extreme events, this summer school is led by GL(EN)² (Glasgow–Lancaster–Edinburgh–Newcastle) Extremes Network and brings PhD students and early-career researchers to the interface of generative AI and extreme value theory. It is an official ECAS course (European Courses in Advanced Statistics).

Partners



THE UNIVERSITY
of EDINBURGH



Universität
Zürich ^{UZH}



University College Dublin
An Coláiste Ollscoile, Baile Átha Cliath



PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE



THE UNIVERSITY of EDINBURGH
Generative AI Laboratory



THE UNIVERSITY of EDINBURGH
Edinburgh Centre for Financial Innovations



THE UNIVERSITY of EDINBURGH
Centre for Investing Innovation

In partnership with
ciabrdn



LEVERHULME
TRUST



Further Reading

In addition to the materials made available by the instructors, we recommend selected parts of the *Handbook of Statistics of Extremes* (<https://extremestats.github.io/>) and parts of the recent special issue on *Bridging Heavy Tails and AI* (<https://edin.ac/4akncn7>).

Day 1: Tuesday

8 September

Theme: *Foundations and Preparations for Generative Modelling*

Moderator: Luca Trapin

Time	Session	Lead
09:00–09:25	Registration & Welcome Coffee	Organisers
09:25–09:30	<i>Welcome & Overview</i>	Chair
09:30–11:00	<i>Introducing Extreme Value Theory</i>	Miguel de Carvalho
11:00–11:30	Coffee break	
11:30–12:30	<i>On Neural Generative Models</i>	Miguel de Carvalho
12:30–14:00	Lunch	
14:00–15:00	<i>Probabilistic Aspects of Diffusions</i>	Sotirios Sabanis
15:00–15:30	Coffee break	
15:30–17:00	<i>Practical: Neural Generative Models in Practice</i>	Johnny Lee

Day 2: Wednesday

9 September

Theme: *Generative Modelling for Extremes and Risk*

Moderator: TBA

Time	Session	Lead
09:30–10:30	<i>GAN-based Generative Approaches for Extremes</i>	Stéphane Girard (online)
10:30–11:00	Coffee break	
11:00–12:00	<i>GAN-based Modelling for Operational Risk</i>	Luca Trapin
12:00–13:30	Lunch	
13:30–15:30	<i>Practical: Generative Modelling for Extremes and Risk</i>	Jordan Richards
15:30–16:00	Coffee break	
16:00–16:10	Group photo	
16:10–17:10	<i>Open Practice & Office Hours</i>	Instructors

Day 3: Thursday

10 September

Theme: *Multivariate Extremes, Cascades, and Simulation*

Moderator: Giorgia Ramponi

Time	Session	Lead
09:30–10:30	<i>Cascades, Multivariate Extremes, and Transformers</i>	Miguel de Carvalho
10:30–11:00	Coffee break	
11:00–12:00	<i>Multivariate Angular Simulation</i>	Emma Simpson
12:00–13:30	Lunch	
13:30–15:30	<i>Transformed-based Approaches</i>	Clemente Ferrer
15:30–16:00	Coffee break	
16:00–17:00	<i>Open Practice & Office Hours</i>	Instructors

Day 4: Friday

11 September

Theme: *Other Frontier Topics*

Moderator: Emma Simpson

Time	Session	Lead
09:30–10:30	<i>Reinforcement Learning</i>	Giorgia Ramponi
10:30–11:00	Coffee break	
11:00–12:00	<i>Pre-trained Extreme Value Modelling</i>	Jordan Richards
12:00–12:10	<i>Final Announcements</i>	Chair
12:10–12:20	<i>Closing Remarks</i>	Chair
12:20–13:50	Lunch	

† Lectures: Lecture Theatre B; practical sessions: Teaching Studio 3217.